CORROSION — Exam Cheat Sheet (Lectures 1–6) · 25 °C unless noted · potentials vs SHE · anodic i > 0, cathodic i < 0

1 · Basics & Constants

Corrosion = electrochemical degradation of metal: **anodic** oxidation $M \rightarrow M^{n+} + ne^-$ + **cathodic** reduction, coupled through the metal (e^-) and the electrolyte (ions). **All 4 needed:** anode, cathode, electrolyte, electronic path → remove one = no corrosion.

Cost ≈ 3–4 % of GDP (NACE IMPACT: ~US$2.5 trillion/yr worldwide); 15–35 % of it is avoidable with existing practice.

Cathodic reactions (E° in V)

Acid H_2 evolution	$2H^+ + 2e^- \rightarrow H_2$	0.000
Neutral/basic H_2	$2H_2O + 2e^- \rightarrow H_2 + 2OH^-$	−0.828
O_2 red., acid	$O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$	+1.229
O_2 red., neutral/basic	$O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$	+0.401
Metal-ion reduction	$Fe^{3+} + e^- \rightarrow Fe^{2+}$	+0.771
Metal deposition	$Cu^{2+} + 2e^- \rightarrow Cu$	+0.337

Constants

$F = 96\,485$ C/mol e⁻ · $R = 8.314$ J/mol·K · $T = 298.15$ K

$\frac{RT}{F} = 0.0257$ V · $\frac{2.303RT}{F} = 0.0592$ V · 1 A = 1 C/s · $T(\mathrm{K}) = T(^\circ\mathrm{C}) + 273.15$

2 · Faraday's Law & Corrosion Rate

$$m = \frac{I\,t\,M}{n\,F} \qquad r = \frac{m}{tA} = \frac{i\,M}{nF}$$

$$\mathrm{CR\ (penetration\ rate)} = \frac{i\,M}{nF\rho}$$

$i = I/A$ (current density), M = atomic mass (g/mol), n = e⁻ per atom, ρ = density (g/cm³).

Watch units: A/cm², cm/s → ×3.156×10⁷ s/yr.

Unit	From current, i in μA/cm²	From mass loss W
mm/y	$3.27\times10^{-3}\,\frac{iM}{n\rho}$	$\frac{87.6\,W}{\rho A t}$ (mg, cm², h)
mpy	$0.129\,\frac{iM}{n\rho}$	$\frac{534\,W}{\rho A t}$ (mg, in², h)

1 mpy = 0.0254 mm/y · mdd = mg dm⁻² day⁻¹ · **Fe: 1 μA/cm² ≈ 0.0116 mm/y ≈ 0.46 mpy**.

Alloy equivalent weight: $EW = \left(\sum \frac{f_i n_i}{M_i}\right)^{-1}$ (f_i = mass fraction) → use EW in place of M/n.

Metal	M	n	ρ	Metal	M	n	ρ
Fe	55.85	2	7.87	Al	26.98	3	2.70
Zn	65.38	2	7.13	Mg	24.31	2	1.74
Cu	63.55	2	8.96	Cr	52.00	3	7.19
Ni	58.69	2	8.90	Pb	207.2	2	11.34

Rough scale: <0.02 mm/y excellent · 0.02–0.1 good · 0.1–0.5 fair · 0.5–1 poor · >1 unacceptable.

3 · Thermodynamics

$$\Delta G = -nFE \qquad \Delta G^\circ = -nFE^\circ = -RT\ln K$$

$$E_{\mathrm{cell}} = E_{\mathrm{cathode}} - E_{\mathrm{anode}}\ \text{(both as reductions)}$$

Spontaneous ⇔ $\Delta G < 0$ ⇔ $E_{\mathrm{cell}} > 0$ ⇔ E_{red} of oxidant > E_{eq} of metal. Thermodynamics says *if*, never *how fast*.

$\Delta G^\circ_{rxn} = \sum \Delta G^\circ_f(\mathrm{prod}) - \sum \Delta G^\circ_f(\mathrm{react})$; elements & H^+, e^-: $\Delta G^\circ_f = 0$. Get $E^\circ = -\Delta G^\circ/nF$.

Nernst equation

For $a\,\mathrm{Ox} + ne^- \rightleftharpoons b\,\mathrm{Red}$:

$$E = E^\circ - \frac{RT}{nF}\ln\frac{a_{\mathrm{Red}}^b}{a_{\mathrm{Ox}}^a} = E^\circ - \frac{0.0592}{n}\log\frac{a_{\mathrm{Red}}^b}{a_{\mathrm{Ox}}^a}$$

Activities: pure solid, H_2O = 1; gas = p (atm); ion ≈ molarity. Metal "corroding" when $[M^{n+}] \geq 10^{-6}$ M.

M^{n+}/M: $E = E^\circ + \frac{0.0592}{n}\log[M^{n+}]$ (Fe at 10^{-6}: −0.617 V)

H^+/H_2: $E = -0.0592\,\mathrm{pH} - 0.0296\log p_{H_2}$

O_2/H_2O: $E = 1.229 - 0.0592\,\mathrm{pH} + 0.0148\log p_{O_2}$

Concentration cell: same metal, different $[M^{n+}]$ → dilute side = anode; $E_{cell} = \frac{0.0592}{n}\log\frac{c_{high}}{c_{low}}$. **Differential aeration:** low-O_2 region = **anode** (crevices, under deposits, waterline).

Standard EMF series (E°, V vs SHE)

Au^{3+}/Au +1.50	Pt^{2+}/Pt +1.19	Ag^+/Ag +0.799
Cu^{2+}/Cu +0.337	H^+/H_2 0.000	Pb^{2+}/Pb −0.126
Sn^{2+}/Sn −0.136	Ni^{2+}/Ni −0.250	Fe^{2+}/Fe −0.440
Cr^{3+}/Cr −0.744	Zn^{2+}/Zn −0.763	Ti^{2+}/Ti −1.63
Al^{3+}/Al −1.662	Mg^{2+}/Mg −2.363	Na^+/Na −2.71

EMF series: pure metals, unit activity, equilibrium, thermodynamic. **Galvanic series:** measured E_{corr} of real metals/alloys in a given environment (e.g. seawater), includes active vs passive states → use it for galvanic-couple prediction.

Reference electrodes (add to convert to SHE)

Electrode	V vs SHE
SCE (sat. calomel, Hg/Hg_2Cl_2)	+0.241
Ag/AgCl (sat. KCl)	+0.197
$Cu/CuSO_4$ (CSE, soil)	+0.316

$E_{\mathrm{SHE}} = E_{\mathrm{ref}} + E_{\mathrm{ref\ vs\ SHE}}$ e.g. −0.500 V_{SCE} = −0.259 V_{SHE}.

4 · Pourbaix (E–pH) Diagrams

General half-reaction: $aA + mH^+ + ne^- \rightleftharpoons bB + cH_2O$

$$E = E^\circ - \frac{0.0592}{n}\log\frac{a_B^b}{a_A^a} - \frac{0.0592\,m}{n}\,\mathrm{pH}$$

Line	Reaction type	Example
Horizontal	e^- only, no H^+ ($m = 0$)	$Fe^{2+} + 2e^- \rightarrow Fe$
Vertical	H^+ only, no e^- ($n = 0$): hydrolysis/precip., $\log K$ fixes pH	$Fe^{2+} + 2H_2O \rightarrow Fe(OH)_2 + 2H^+$
Sloped	both; slope = $-0.0592\,m/n$	$Fe_2O_3 + 6H^+ + 2e^- \rightarrow 2Fe^{2+} + 3H_2O$

Water lines (dashed): **(a)** $E = -0.0592\,\mathrm{pH}$ — below, H_2 evolves; **(b)** $E = 1.229 - 0.0592\,\mathrm{pH}$ — above, O_2 evolves. Between them water is stable.

Regions: Immunity (metal stable) · **Corrosion** (soluble ion >10^{-6} M) · **Passivation** (solid oxide/hydroxide; *may* protect).

Use: metal line below line (a) → corrodes in deaerated acid with H_2 evolution; below line (b) → O_2 can corrode it. Protection routes: lower E into immunity (cathodic prot.), raise E into passivity (anodic prot.), change pH.

Limitations: equilibrium only — **no kinetics/rates**; pure metal, pure water at 25 °C; ignores aggressive ions (Cl^- → pitting), complexing, alloys; "passive" region assumes the film is protective; boundary choice (10^{-6} M) is arbitrary.

5 · Electrode Kinetics

Overpotential $\eta = E - E_{eq}$: $\eta > 0$ anodic, $\eta < 0$ cathodic. **Polarization** = shift from E_{eq} due to net current. Types: **activation** (charge-transfer slow), concentration (not on exam), **ohmic/IR** ($\eta_\Omega = iR$).

Exchange current density i_0: equal & opposite rates at E_{eq} ($\vec{i} = |\overleftarrow{i}| = i_0$, net 0). Depends on reaction, electrode surface, T, concentration; $i_0 \propto e^{-\Delta G^\ddagger/RT}$. H_2 on: Pt ~10^{-3}, Ni ~$10^{-5.6}$, Fe ~10^{-6}, Cu ~10^{-7}, Zn ~10^{-11}, Pb/Hg ~10^{-12}–10^{-13} A/cm².

Butler–Volmer

$$i_{net} = i_0\left[e^{\alpha_a nF\eta/RT} - e^{-\alpha_c nF\eta/RT}\right]$$

$\alpha_a + \alpha_c = 1$ (symmetry factor, usually 0.5)

High field ($|\eta| \gtrsim 50$–100 mV) → one term dominates → **Tafel**:

$$\eta_a = \beta_a\log\frac{i}{i_0}, \quad \beta_a = \frac{2.303RT}{\alpha_a nF}$$

$$\eta_c = \beta_c\log\frac{|i|}{i_0}, \quad \beta_c = -\frac{2.303RT}{\alpha_c nF}$$

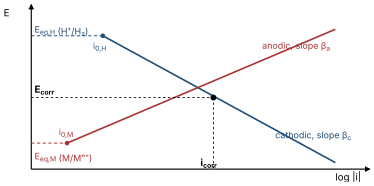
$\alpha = 0.5$, $n = 1$: $|\beta| \approx 0.118$ V/decade (~120 mV/dec). Typical 0.03–0.2 V/dec. Plot E vs $\log|i|$ → straight line, slope β, intercept at $\eta = 0$ gives $\log i_0$. Also $\eta = a + b\log i$ with $a = -\beta\log i_0$.

Low field ($|\eta| \lesssim 10$ mV, linearize $e^x \approx 1 + x$): $\eta = \frac{RT}{nF}\frac{i}{i_0}$ → charge-transfer resistance $R_{ct} = \frac{RT}{nFi_0}$.

6 · Mixed Potential Theory

Wagner–Traud hypotheses: (1) any electrochemical reaction can be divided into ≥2 partial oxidation & reduction reactions; (2) no net accumulation of charge: $\sum i_{ox} = \sum |i_{red}|$ (with areas: $\sum I_a = \sum |I_c|$).

Corrosion potential E_{corr} (mixed / open-circuit potential): where total anodic = total cathodic current. i_{corr} = anodic current of the metal there. $E_{eq,M} < E_{corr} < E_{eq,cath}$; E_{corr} is **not** an equilibrium potential (net reaction proceeds).



Evans diagram: intersection of anodic (metal) & cathodic Tafel lines.

Solving E_{corr}, i_{corr} (set the two Tafel lines equal; β_c negative)

$$E_{corr} = E_{eq,M} + \beta_a\log\frac{i_{corr}}{i_{0,M}} = E_{eq,H} + \beta_c\log\frac{i_{corr}}{i_{0,H}}$$

$$\log i_{corr} = \frac{N}{\beta_a - \beta_c}, \quad \text{where}$$

$$N = E_{eq,H} - E_{eq,M} + \beta_a\log i_{0,M} - \beta_c\log i_{0,H}$$

Recipe: (1) Nernst → E_{eq} of each reaction at given pH/conc.; (2) write Tafel lines; (3) equate → i_{corr}; (4) back-substitute → E_{corr}; (5) Faraday → CR.

What the Evans diagram predicts

- **Larger i_0 of cathodic reaction** (e.g. Pt vs Hg surface) → higher i_{corr}, nobler E_{corr}.
- **Zn vs Fe in acid:** Zn more active (E° lower) but $i_{0,H}$ on Zn ≪ on Fe → Zn can corrode **slower**. Thermodynamic driving force ≠ rate.
- **Adding oxidizer** (O_2, Fe^{3+}): extra cathodic line; total cathodic = sum of currents at each E → E_{corr} ↑, i_{corr} ↑. Fe^{3+} also reduces i of H_2 evolution at new E_{corr}.
- **Lower pH** → $E_{eq,H}$ rises (−59 mV/pH) → i_{corr} ↑.
- **Several anodic metals**: sum anodic currents; each metal has its own i at the common E.

Applied (external) current

$i_{app} = i_a - |i_c|$ at potential E. **Anodic polarization** ($E > E_{corr}$): $i_{app,a} = i_a - |i_c|$. **Cathodic** ($E < E_{corr}$): metal dissolution ↓ (basis of **cathodic protection**: polarize to $E_{eq,M}$ → $i_a \rightarrow i_{0,M}$, full protection). Measured curve is curved near E_{corr} and only follows the Tafel line when $|E - E_{corr}| \gtrsim 50$–100 mV.

Galvanic coupling

Two metals electrically connected in same electrolyte → common E_{couple} between the two E_{corr}'s. Active metal: E raised → corrodes **faster**; noble metal: E lowered → corrodes slower (cathodically protected). Condition, with areas:

$$\sum I_a = \sum |I_c| \ \Rightarrow$$

$$i_{a,M}A_M = |i_{c,M}|A_M + |i_{c,N}|A_N$$

(M = active metal, N = noble metal, cathodic rxn on both)

Area effect: $i_a \propto A_{cath}/A_{anode}$ (cathodic control) → **small anode / large cathode = worst** (e.g. steel rivets in Cu plate). Plot currents (not densities) when areas differ: shift each line by $\log A$.

Galvanic series in seawater (active → noble): Mg · Zn · galv. steel · Al alloys · Cd · mild steel / cast iron · SS 304/316 (active) · Pb · Sn · Ni (active) · brass · Cu · bronze · Cu–Ni · Monel · Ni (passive) · SS 304/316 (passive) · Ag · Ti · graphite · Au · Pt.

Prevention: pick metals close in series; insulate joints; avoid small-anode geometry; coat the *cathode* (never only the anode); sacrificial anode.

7 · Polarization Measurements

3-electrode cell: WE (sample), RE (+ Luggin capillary → minimize IR), CE (Pt/graphite, carries current); **potentiostat** controls E_{WE} vs RE, measures I through CE. Slow scan (ASTM G5: 0.6 V/h ≈ 0.17 mV/s). $E_{meas} = E_{true} + iR_s$.

Tafel extrapolation: extrapolate linear Tafel region(s) of polarization curve back to E_{corr} → intersection gives i_{corr}. Needs ≥1 decade of linear region; destructive (large η).

Linear polarization resistance (non-destructive, ±10–20 mV about E_{corr}):

$$R_p = \left(\frac{\partial\Delta E}{\partial i_{app}}\right)_{\Delta E\rightarrow 0} \qquad i_{corr} = \frac{B}{R_p}$$

$$B = \frac{\beta_a|\beta_c|}{2.303(\beta_a + |\beta_c|)}$$

Stern–Geary. R_p in Ω·cm² → i_{corr} in A/cm². With $\beta_a = |\beta_c| = 0.12$ V: $B \approx 0.026$ V. One β → ∞ (e.g. diffusion): $B = \beta/2.303$. Remember to subtract solution resistance: $R_p = R_{meas} - R_s$.

8 · Passivity (active–passive metals)

Thin (1–10 nm) protective oxide film (Fe, Cr, Ni, Ti, Al, SS). Anodic curve: active Tafel region → peak at E_{pp} (primary passivation potential) with i_{crit} (critical current density) → current drops to i_{pass} (10^3–10^6× lower) → **transpassive** (film dissolves/O_2 evolution) or E_b/E_{pit} breakdown with Cl^-. Flade potential ≈ E_{pp} on reactivation.

Cathodic line intersects anodic curve in: (1) active only → active corrosion; (2) active + passive (3 intersections, middle unstable) → unstable, may activate if film damaged; (3) passive only → **spontaneous passivation**, needs $|i_c(E_{pp})| > i_{crit}$.

Oxidizer ↑ on passive metal → can *lower* corrosion (to i_{pass}) — opposite to active metals; too much → transpassive. Easier passivation: lower i_{crit}, more active E_{pp} (add Cr ≥ 10.5–12 % for SS, Mo, Ni). **Anodic protection**: hold E in passive range. PREN = %Cr + 3.3 %Mo + 16 %N.

9 · Forms of Corrosion (Fontana 8)

Uniform · Galvanic · Crevice · Pitting · Intergranular (sensitization: $Cr_{23}C_6$ at GB, 425–815 °C) · Selective leaching (dezincification) · Erosion-corrosion (cavitation, fretting) · Stress-corrosion cracking (+ H damage). Prevention: material selection, design, coatings, inhibitors, cathodic/anodic protection, environment change.

Exam traps: convert reference scales before comparing · use $|i|$ in logs, β_c negative · i vs I (×area) · n per atom · cm vs mm vs in · E° is reduction potential — flip sign only conceptually, not in $E_{cell} = E_c - E_a$ · thermodynamics ≠ kinetics.